



Department of Artificial Intelligence and Data Science

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: B.Tech, IV & Artificial Intelligence and Data Science & 'A'

Course Code & Title: AL3451 & Machine Learning

Name of the Faculty member (s): Dr. S. Selva Birunda

Innovative Practice Description

- **Unit / Topic:** Unit III /Ensemble Learning
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 7, 8
- **Activity Chosen:** Mind Map
- **Justification:**

A Mind Map activity was conducted to visually organize information into a hierarchy which demonstrated the connections between the various parts of the whole. Through this activity, their understanding of Ensemble Learning and its types is strengthened. It frequently starts with a single thought, depicted as an image in the middle of a blank page, to which related images, words, or word fragments are added. A mind map is a simple tool for naturally generating ideas without worrying about organisation or order.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:**
 1. Instructor explained the Ensemble Learning concepts in classroom.
 2. Based on the discussion, the instructor asked the students to create a mind map.
 3. Each student created a mind map on Bagging, Boosting and Stacking concepts
 4. They identify key subtopics and add supporting details
 5. Students represent the concepts visually including Central theme, Associations, Curved Lines and Keywords.



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- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students felt the mind map activity engaging and helpful in visualizing key concepts
 - ✓ Students appreciated the interactive nature, stating it improved their retention of information

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ Students can think outside the box or from diverse perspectives by participating in this activity
- ✓ Improves understanding and helps in breaking down the complex topics into simpler parts

❖ ***Challenges faced in implementation:***

- ✓ Some students struggled with organizing their ideas effectively

References:

1. <https://www.mindmapping.com/mind-map>
2. <https://www.mindmapping.com/>



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FEEDBACK

Active Learning Best practices: Mind Map Active Learning Technique

Degree, Semester & Branch : B. Tech, IV Semester & AI / DS

Course Code & Title : AL3451 & Machine Learning

Name of the Faculty member (s) : Dr. S. Selva Birunda, ASP / AD

Theme of discussion: Ensemble Learning

Date and Time: 26.03.2025 & 3.45 pm to 4.30 pm

Feedback collected in class and also through online

Feedback Questionnaire:

1. How useful was the mind map paper activity that we did in class today?

Very much useful

Not useful

2. Do you have a clear understanding of the Ensemble Learning concepts?

Excellent

Very Good

Good

Satisfactory

3. Does this mind map activity improve listening and grasping capability?

Excellent

Very Good

Good

Satisfactory

Feedback Analysis:

Attempts: 53 out of 53

-0

Discrimination Index ⓘ

How useful was the mind map activity that we did in class today?

Very Much useful	53 respondents	100 %	✓
Not useful		0 %	

100% answered correctly

Attempts: 53 out of 53

+0.89

Discrimination Index ⓘ

Do you have a clear understanding of the Ensemble Learning concepts?

Excellent	49 respondents	92 %	✓
Very Good	3 respondents	6 %	
Good		0 %	
Satisfactory	1 respondent	2 %	

92% answered correctly

Attempts: 53 out of 53

+0.43

Discrimination Index ⓘ

Does this mind map activity improve listening and grasping capability?

Yes	52 respondents	98 %	✓
No	1 respondent	2 %	

98% answered correctly

S. S. / 8/4/25
Faculty in-charge

mark / 8/4/25
HOD/AD